Gateway PQC Re-Encryption Pipeline Report

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Source: `reencrypt_metrics.csv + firewall_misses.jsonl`

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1. Executive Summary

The gateway re-encrypted **96 benign messages** using `ML-KEM-512` + `ML-DSA-44` (AES-256-GCM). Mean broker transit was **1.385 ms** and mean PQC re-encryption overhead was **0.168 ms**.

Metric	Value
Messages re-encrypted	96
KEM algorithm	`ML-KEM-512`
Signature algorithm	`ML-DSA-44`
PQC mode	Real liboqs
Observation window	17:37:04 → 17:38:02

2. Timing Breakdown

All values in milliseconds (ms). **Broker transit** = `t_recv_ns − t_publish_ns` (includes broker + network latency). **Total gateway** = full handler wall time from `on_message` entry.

Stage	N	Mean	Std	P95	Min	Max
Broker transit (client→gateway)	96	1.385 ms	0.612 ms	2.652 ms	0.905 ms	3.227 ms
KEM encapsulate	96	0.050 ms	0.010 ms	0.064 ms	0.038 ms	0.108 ms
AES-256-GCM encrypt	96	0.028 ms	0.053 ms	0.025 ms	0.018 ms	0.536 ms
Signature sign	96	0.089 ms	0.036 ms	0.167 ms	0.049 ms	0.215 ms
Total re-encrypt (KEM+AES+Sign)	96	0.168 ms	0.072 ms	0.260 ms	0.112 ms	0.739 ms
Total gateway (on_message→done)	96	0.450 ms	0.569 ms	0.512 ms	0.318 ms	5.970 ms

3. Per-Device Timing

Device ID	N	Transit Mean	Transit Std	Reencrypt Mean	Reencrypt Std	Total GW Mean
benign-burst-dc2419	66	1.358 ms	0.574 ms	0.164 ms	0.044 ms	0.394 ms
benign-steady-59be62	30	1.444 ms	0.684 ms	0.176 ms	0.110 ms	0.576 ms

4. Firewall Misses — Attacks That Reached the Gateway

Total missed attacks: 50

These messages had `is_attack=True` in the payload, meaning the attack client injected the label. The LightGBM firewall failed to classify or block the corresponding network flow before it reached the gateway. Use these records to identify model weaknesses.

By Attack Label

Attack Label	Count	Share
mqtt-flood	50	100.0%

By crypto_type Used (Spoofing Pattern)

crypto_type	Count
aes_gcm	50

Last 10 Miss Events

Time	Attack Label	Device	Seq	crypto_type	Payload Preview
17:37:40	mqtt-flood	flood-attacker	14022616	aes_gcm	{"device_id": "flood-attacker", "crypto_type": "aes_gcm", "s
17:37:40	mqtt-flood	flood-attacker	14022617	aes_gcm	{"device_id": "flood-attacker", "crypto_type": "aes_gcm", "s
17:37:41	mqtt-flood	flood-attacker	14022618	aes_gcm	{"device_id": "flood-attacker", "crypto_type": "aes_gcm", "s
17:37:41	mqtt-flood	flood-attacker	14022619	aes_gcm	{"device_id": "flood-attacker", "crypto_type": "aes_gcm", "s
17:37:41	mqtt-flood	flood-attacker	14022620	aes_gcm	{"device_id": "flood-attacker", "crypto_type": "aes_gcm", "s
17:37:41	mqtt-flood	flood-attacker	14022621	aes_gcm	{"device_id": "flood-attacker", "crypto_type": "aes_gcm", "s
17:37:41	mqtt-flood	flood-attacker	14022622	aes_gcm	{"device_id": "flood-attacker", "crypto_type": "aes_gcm", "s
17:37:42	mqtt-flood	flood-attacker	14022623	aes_gcm	{"device_id": "flood-attacker", "crypto_type": "aes_gcm", "s
17:37:42	mqtt-flood	flood-attacker	14022624	aes_gcm	{"device_id": "flood-attacker", "crypto_type": "aes_gcm", "s
17:37:42	mqtt-flood	flood-attacker	14022625	aes_gcm	{"device_id": "flood-attacker", "crypto_type": "aes_gcm", "s

5. Last 10 Re-Encrypted Messages

Device	Seq	Broker Transit	Re-Encrypt	Total GW
benign-burst-dc2419	63	0.999 ms	0.135 ms	0.348 ms
benign-burst-dc2419	64	1.306 ms	0.251 ms	0.509 ms
benign-burst-dc2419	65	0.939 ms	0.221 ms	0.442 ms
benign-burst-dc2419	66	1.093 ms	0.143 ms	0.358 ms
benign-steady-59be62	25	1.443 ms	0.190 ms	0.438 ms
benign-steady-59be62	26	0.996 ms	0.174 ms	0.419 ms
benign-steady-59be62	27	2.709 ms	0.221 ms	0.465 ms
benign-steady-59be62	28	0.981 ms	0.151 ms	0.403 ms
benign-steady-59be62	29	1.103 ms	0.200 ms	0.414 ms
benign-steady-59be62	30	3.156 ms	0.134 ms	0.344 ms

Report generated by *scripts/parse_gateway_reencrypt.py* — MQTT IDS/IPS + PQC Benchmarking Framework.